Project title: EduGenie -Google Gemini Powered Learning Assistant

Phase 4: Project Planning Phase

1. Project Planning Phase:

-The project planning phase focuses on organizing the development activities and deciding the sequence in which the project should be implemented.

-Initially, the educational problem was identified and the main objectives of EduGenie were defined. After that, the required technologies, modules, and development environment were selected.

-The project development was divided into several stages.

Development Plan:

Stage 1 – Problem Identification:

Identify the difficulties faced by students while learning and determine how AI can provide learning assistance.

Stage 2 – Requirement Collection:

Identify the functional and non-functional requirements of the application.

Stage 3 – System Design:

Design the application architecture, workflow, modules, and user interaction.

Stage 4 – Environment Setup:

Install Python, create the project environment, install required packages, and configure the development environment.

Stage 5 – Backend Development:

Develop the FastAPI backend and create the required API endpoints.

Stage 6 – Gemini Integration:

Connect the backend with Google Gemini and configure the required API credentials.

Stage 7 – Frontend Development:

Develop the user interface using HTML, CSS, and JavaScript.

Stage 8 – Module Development:

Implement the required learning modules such as Q&A, quiz, explanation, and summary functions.

Stage 9 – Testing:

Test the application using different inputs and verify the generated outputs.

Stage 10 – Documentation and Demonstration:

Prepare the project documentation, screenshots, sample outputs, and final project demonstration.

-This systematic planning helped us complete the project in an organized manner.